

From Coast to Coast: Comparing How Air Pollution Influences Asthma in Two States

Sakina Tariq, sztariq@albany.edu

CEHC - University at Albany, State University of New York



INTRODUCTION

Asthma is influenced by both environmental and geographic factors, including exposure to fine particulate matter (PM2.5). Using the Asthma Equity Explorer, this study compares PM levels and asthma prevalence across counties in New York and California. By examining how this relationship differs between states, we aim to better understand how air quality contributes to asthma patterns in different regions.

DATASET SELECTION & DESCRIPTIVE STATISTICS

Data Source: Asthma Equity Explorer (Allergy & Asthma Network)

Variables Used:

- Asthma prevalence (%)
- PM2.5 concentration (micrograms per cubic meter)

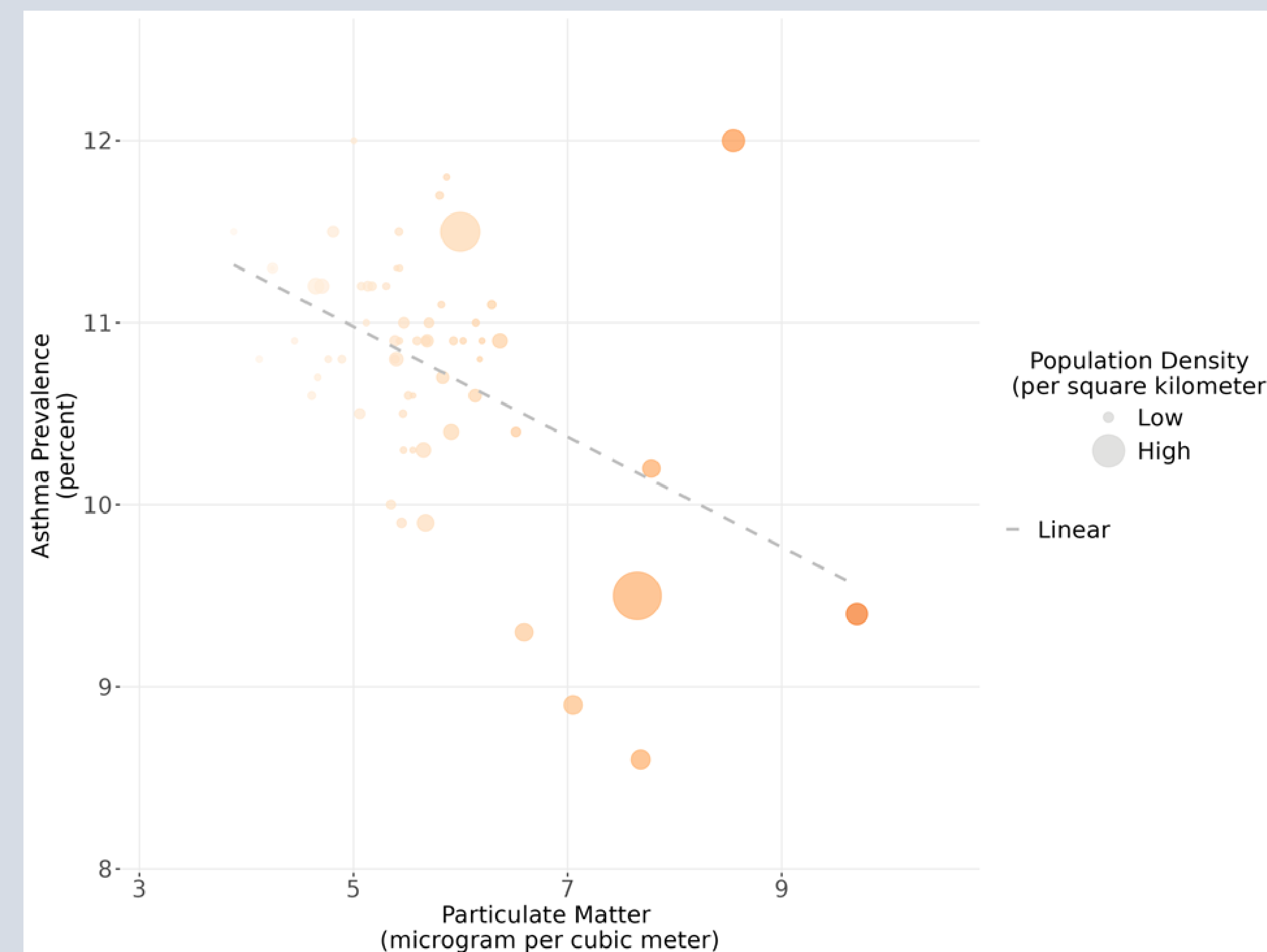
Geographic Units:

- **New York:** 62 counties
- **California:** 58 counties

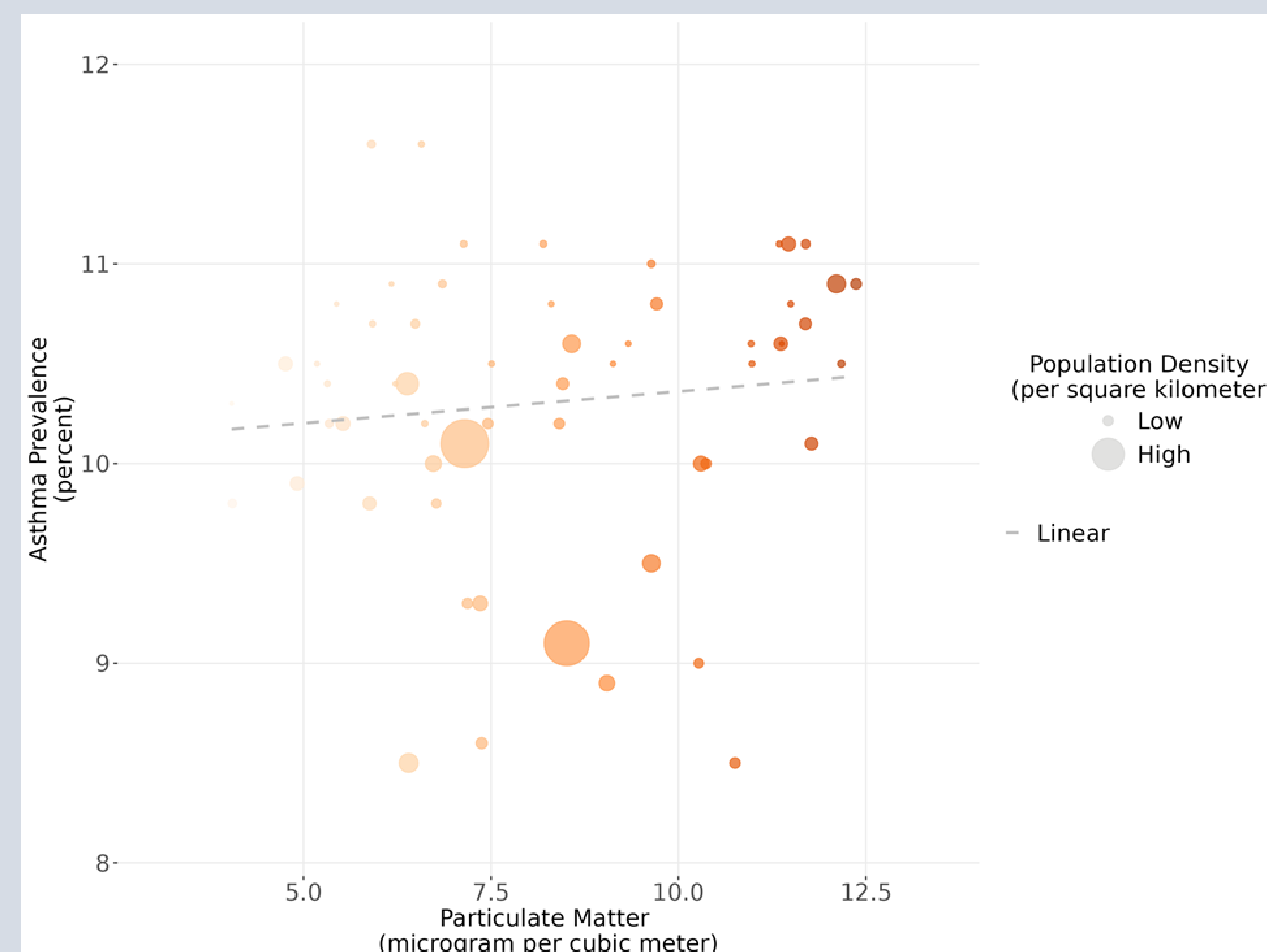
State Averages/Mean:

- **New York:** Asthma 10.78%, PM2.5 = $5.67 \mu\text{g}/\text{m}^3$
- **California:** Asthma 10.31%, PM2.5 = $8.24 \mu\text{g}/\text{m}^3$

EXPLORATORY DATA ANALYSIS



“With the factor you have specified, Particulate Matter is significantly associated with Asthma Prevalence for NY. The relationship is negative and weak, considering the 62 counties within NY.”
(Asthma equity explorer)



“With the factor you have specified, Particulate Matter is not significantly associated with Asthma Prevalence for CA.” (Asthma equity explorer)

RECOMMENDATIONS/FUTURE RESEARCH

- Explore why the PM2.5 asthma relationship differs between New York and California, including the potential influence of geography, terrain, and climate.
- Investigate whether natural features such as mountain ranges, valleys, or coastal airflow patterns affect how pollution spreads and impacts respiratory health.
- Examine how county-level differences in population size and density contribute to asthma variation beyond pollution levels.
- Assess whether using individual-level samples or household-level data (rather than county averages) would change the observed relationships between PM2.5 and asthma in each state.

REFERENCES

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- California Air Resources Board. Asthma & Air Pollution | California Air Resources Board. (n.d.). <https://ww2.arb.ca.gov/resources/asthma-and-air-pollution>
- Mananga, E. S., Lopez, E., Diop, A., Dongomale, P. J., & Diane, F. (2023, November 29). *The impact of the air pollution on Health in New York City*. Journal of public health research. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10687960/>